

EARTH: Accelerating Spatiotemporal Network K-function-based Analytics

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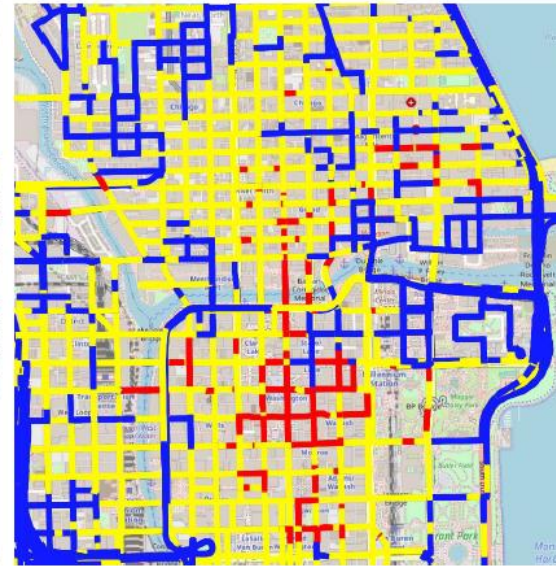
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Is a Hotspot a Hotspot?



(a) Chicago crime data points

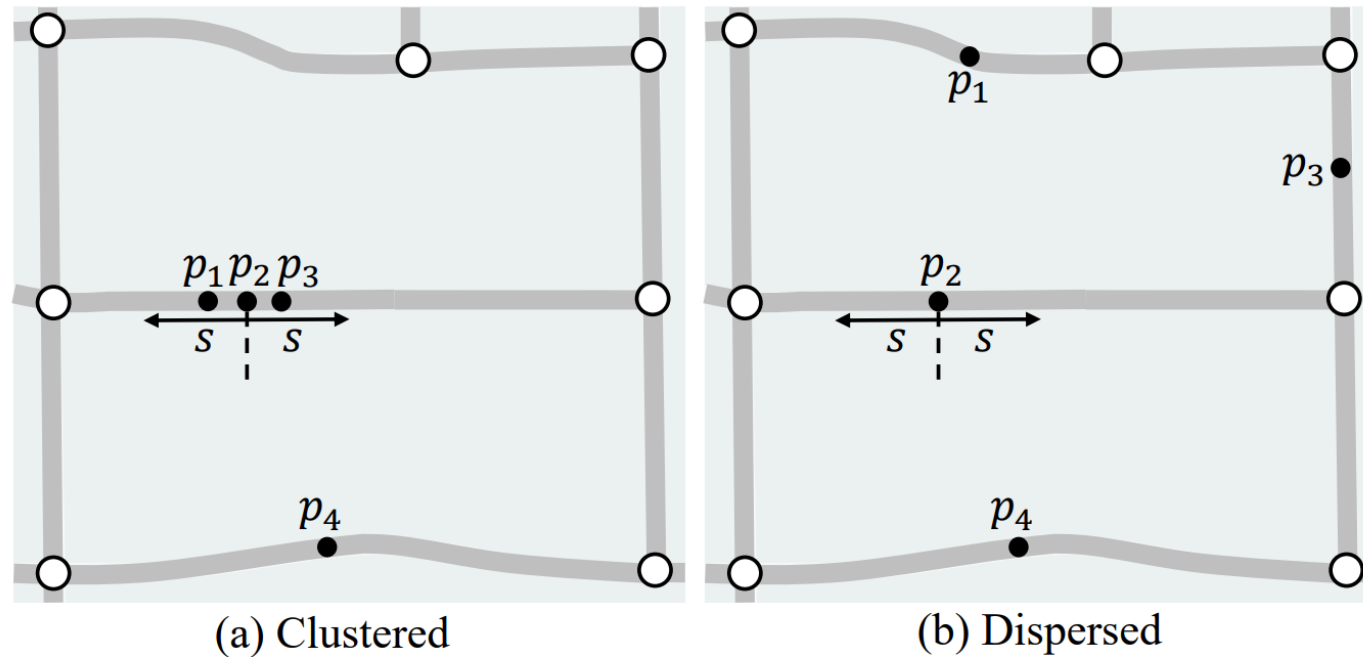


(b) NKDV

- Do you think that the red part is really a hotspot?
- How do you tune the parameters in some statistical/clustering models?

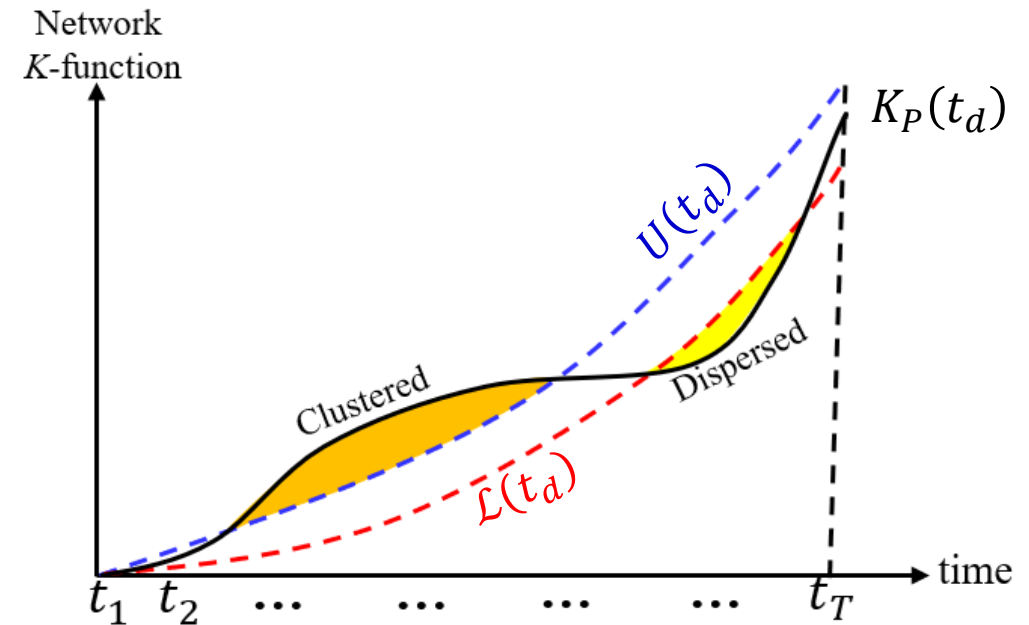
What is Network K-function?

- Count the number of data points that are within the spatial threshold s from each data point.



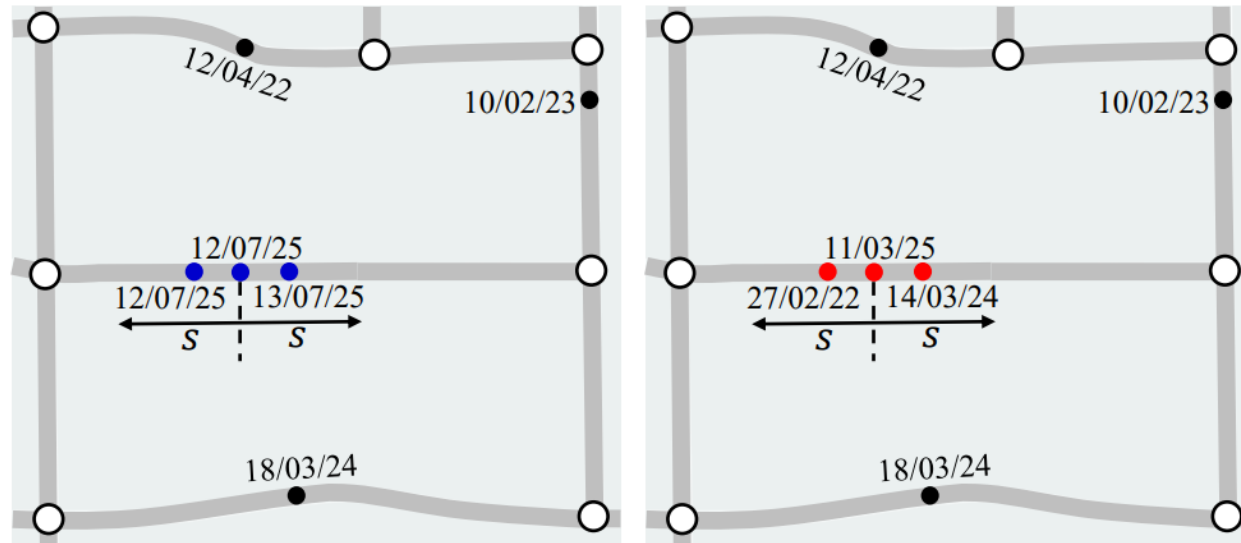
What is Network K-function Plot?

- Domain experts need to determine whether the hotspots/clusters of a location dataset are meaningful.
- Generate multiple random datasets with the same size as the original dataset.
- Compute the network K-function values for the original dataset and random datasets with respect to multiple thresholds.



Weakness of Network K-function

- Ignore the timestamp of each data point.

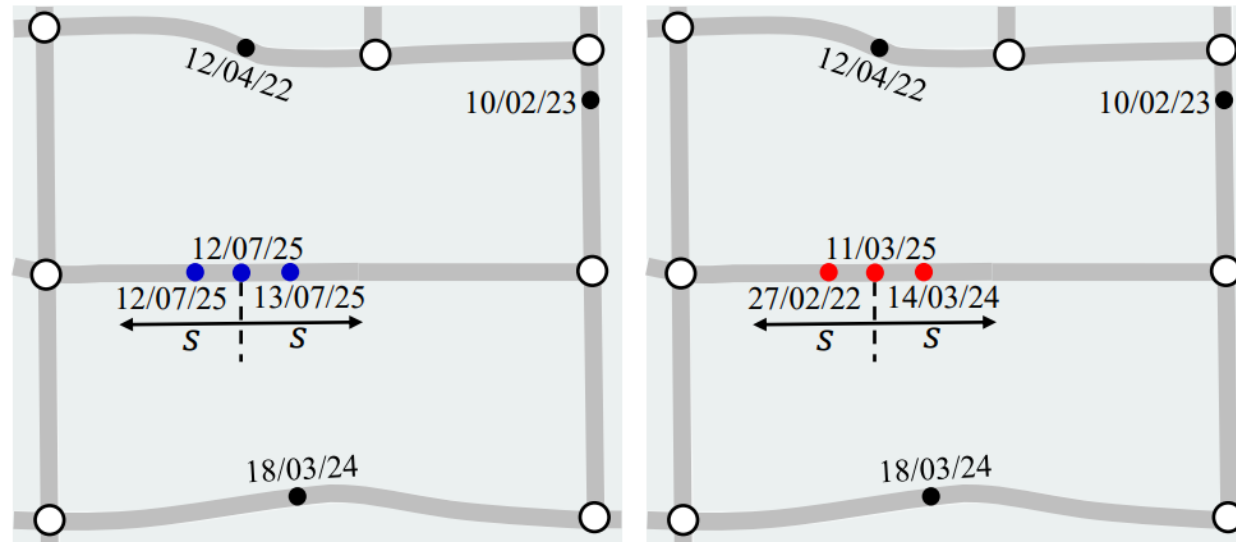


- Many spatiotemporal events depend on time.

Overview of Spatiotemporal Network K-function

- Given a road network $G = (V, E)$, a spatiotemporal dataset $P = \{(p_1, \tau_{p_1}), (p_2, \tau_{p_2}), \dots, (p_n, \tau_{p_n})\}$, a spatial threshold s , and a temporal threshold t , we need to compute $K_P(s, t)$.

$$K_P(s, t) = \sum_{(p_i, \tau_{p_i}) \in P} \sum_{\substack{(p_j, \tau_{p_j}) \in P \\ j \neq i}} \mathbb{I}(d_G(p_i, p_j) \leq s, d(\tau_{p_i}, \tau_{p_j}) \leq t)$$

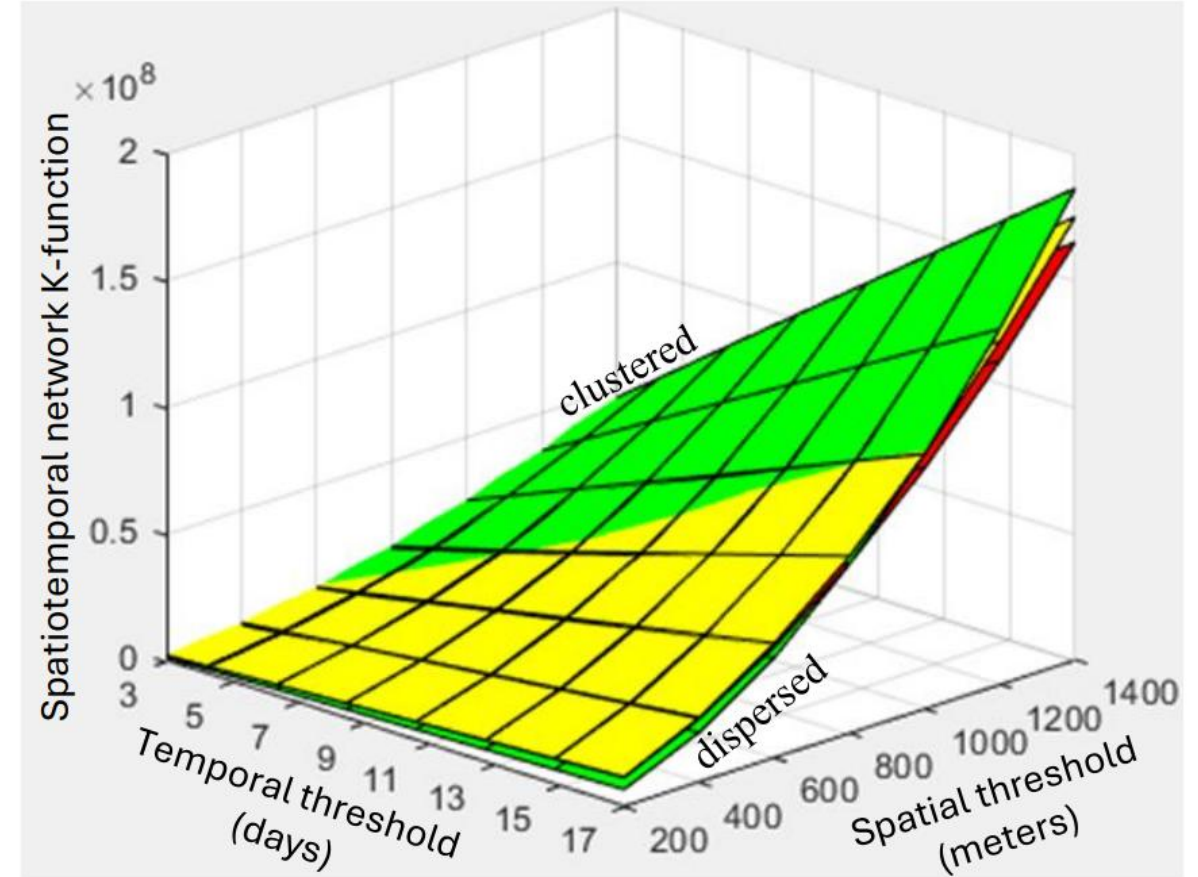


Spatiotemporal Network K-function Plot

- Generate L datasets, $R_1, R_2, \dots, R_L$, with the same size n as P .
- Choose M spatial thresholds, $s_1, s_2, \dots, s_M$, and T spatial thresholds, $t_1, t_2, \dots, t_T$.
- Compute $K_P(s_\alpha, t_\beta)$ (green plane), $\mathcal{L}(s_\alpha, t_\beta)$ (red plane), $\mathcal{U}(s_\alpha, t_\beta)$ (yellow plane), where

$$\mathcal{L}(s_\alpha, t_\beta) = \min \left(K_{R_1}(s_\alpha, t_\beta), K_{R_2}(s_\alpha, t_\beta), \dots, K_{R_1}(s_\alpha, t_\beta), K_{R_L}(s_\alpha, t_\beta) \right)$$

$$\mathcal{U}(s_\alpha, t_\beta) = \max \left(K_{R_1}(s_\alpha, t_\beta), K_{R_2}(s_\alpha, t_\beta), \dots, K_{R_1}(s_\alpha, t_\beta), K_{R_L}(s_\alpha, t_\beta) \right)$$



Spatiotemporal Network K-function is Slow!

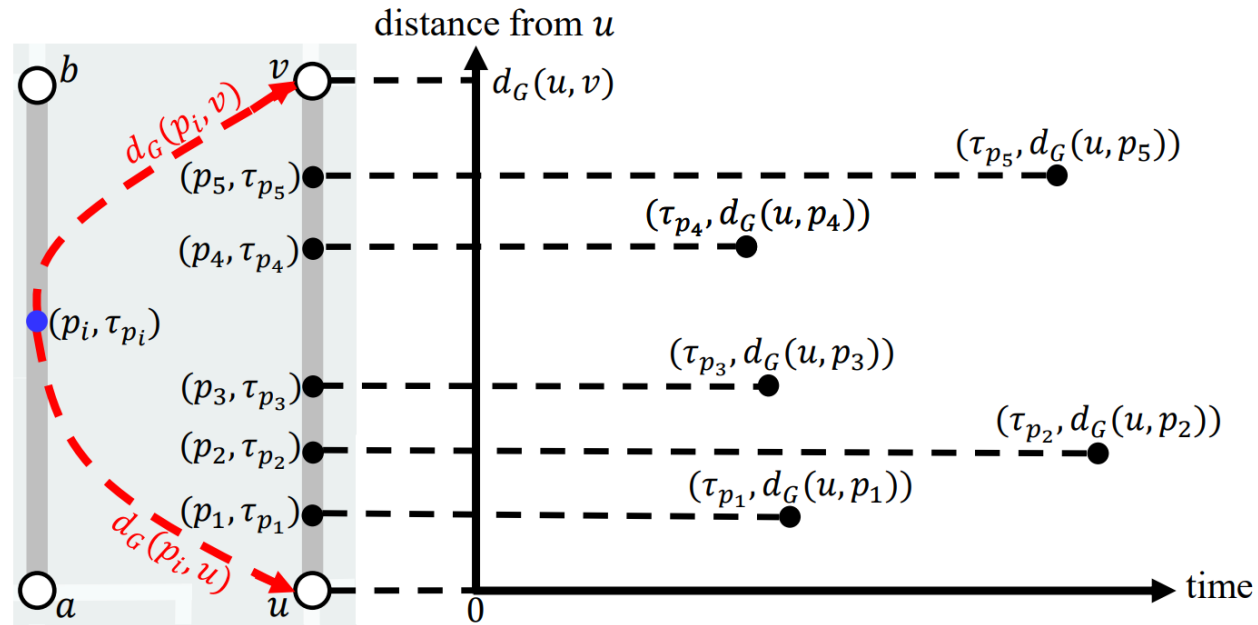
- Generate a spatiotemporal network K-function takes $O(|E|T_{SP} + n^2)$ time. ☹️
- Generate a spatiotemporal network K-function plot takes $O(LMT(|E|T_{SP} + n^2))$. ☹️
- Infeasible even when $n = 1000000$. ☹️

Our Contributions

Problem	Method	Time complexity	Space complexity
Spatiotemporal network K -function	RQS [54]	$O(n(\mathcal{T}_{SP} + n))$	$O(\mathcal{S}_{SP} + n)$
	SPS [55]	$O(E \mathcal{T}_{SP} + n^2)$	
	EAR (Section 4.1)	$O(E \mathcal{T}_{SP} + n E \log(\frac{n}{ E }) + n \log n)$	$O(\mathcal{S}_{SP} + n)$ (Section 4.4)
Spatiotemporal network K -function plot	RQS [54]	$O(LMTn(\mathcal{T}_{SP} + n))$	$O(\mathcal{S}_{SP} + Ln + LMT)$
	SPS [55]	$O(LMT(E \mathcal{T}_{SP} + n^2))$	
	EAR (Section 4.1)	$O(E \mathcal{T}_{SP} + LMTn E \log(\frac{n}{ E }) + Ln \log n)$	$O(\mathcal{S}_{SP} + Ln + LMT)$ (Section 4.4)
	MTS (Section 4.2)	$O(E \mathcal{T}_{SP} + Ln^2 + LMTn E)$	
	EARTH (Section 4.3)	$O(E \mathcal{T}_{SP} + \min(LMTn E \log(\frac{n}{ E }) + Ln \log n, Ln^2 + LMTn E))$	

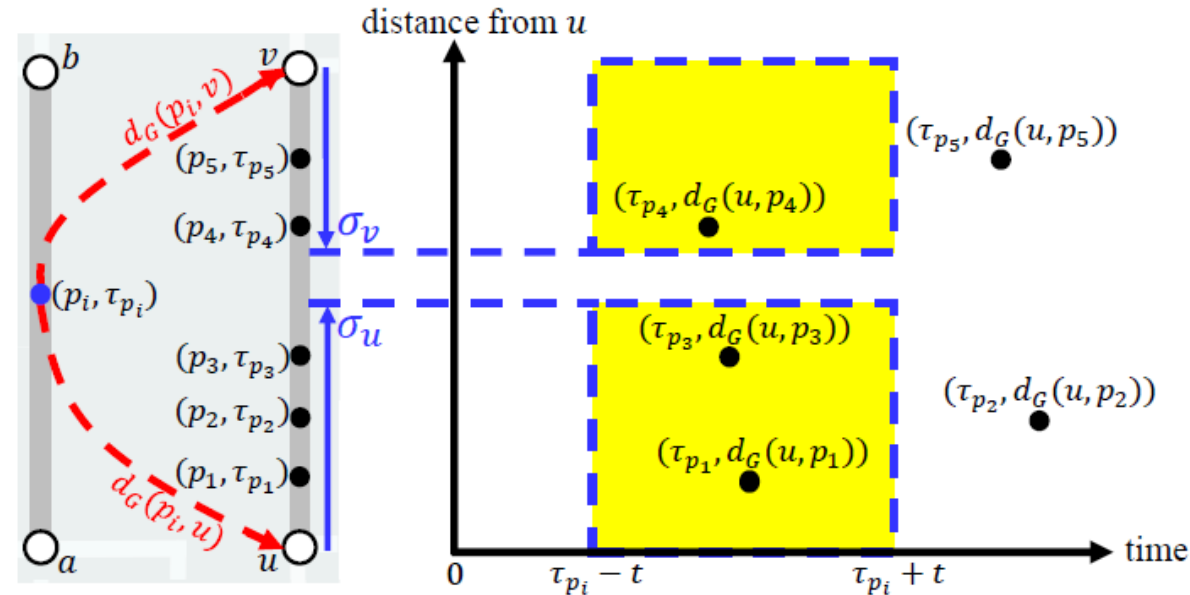
- Develop new methods that can theoretically **reduce the time complexity with the same space complexity**. 😊
- Achieve up to 19.04x speedups compared with existing methods. 😊
- Have integrated these methods into the python library, called FastNSA. 😊

Core Idea of Our Methods: Mapping



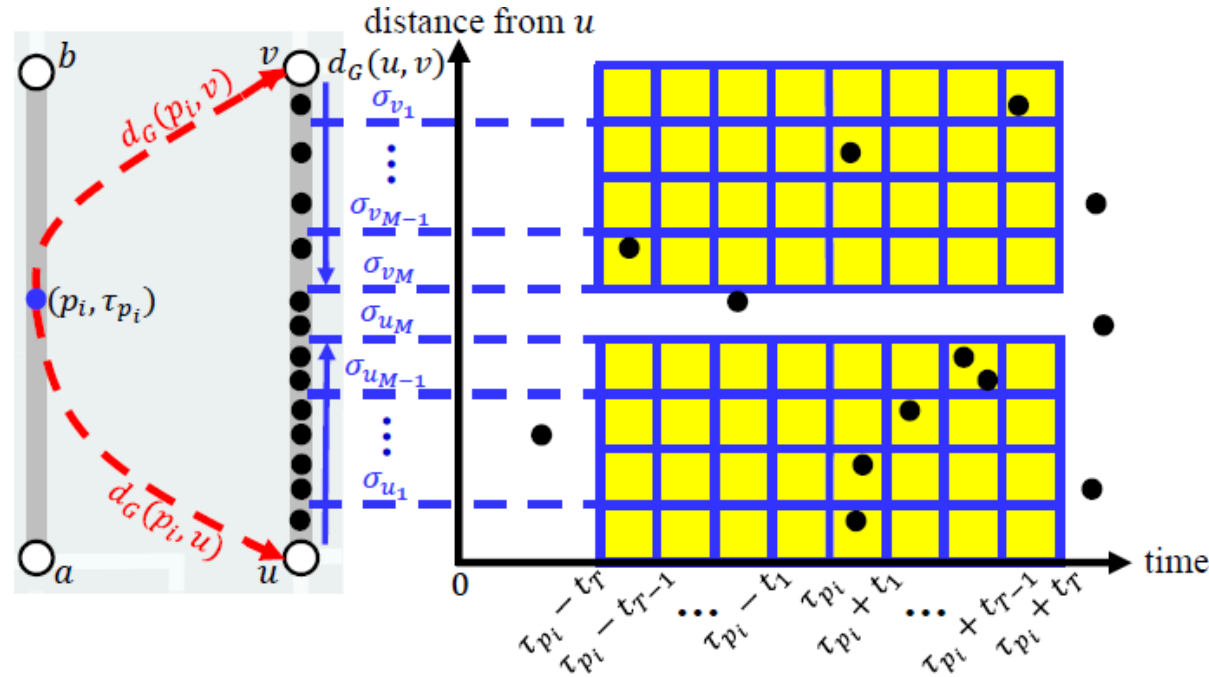
- Map all data points on the edge $e = (u, v)$ to the two-dimensional (time, distance)-plane.

Edge Augmentation with Range-tree (EAR)



- Counting data points for each (p_i, τ_{p_i}) is equivalent to the 2-D orthogonal range counting problem.
- Build range-tree for each edge in advance and use it for each (p_i, τ_{p_i}) .
 - $O(|P(e)|)$ time $\rightarrow O(\log|P(e)|)$ time. ☺

Multi-Threshold Sharing (MTS)



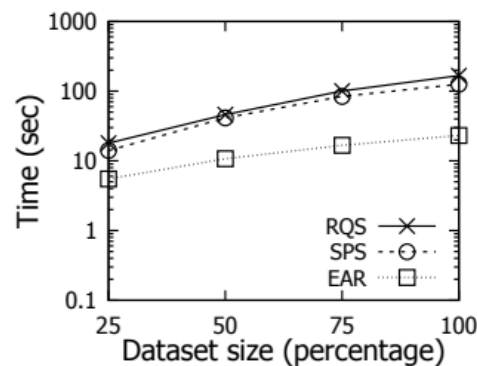
- Counting data points with M spatial thresholds and T temporal thresholds for each (p_i, τ_{p_i}) is equivalent to counting data points in the grid structure.
- Build prefix-sum structure for each edge in advance and use it for each (p_i, τ_{p_i}) with multiple thresholds.
 - $O(MT|P(e)|)$ time $\rightarrow O(MT + |P(e)|)$ time. ☺

EARTH

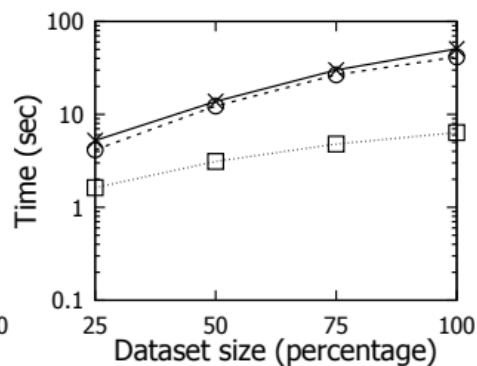
- EARTH selects either EAR or MTS for each edge based on the cost function.
 - $\text{EARTH} = \text{EAR} + \text{MTS}$
 - Achieves the smallest time complexity. 😊
- For details, please refer to our paper.

Experimental Evaluation

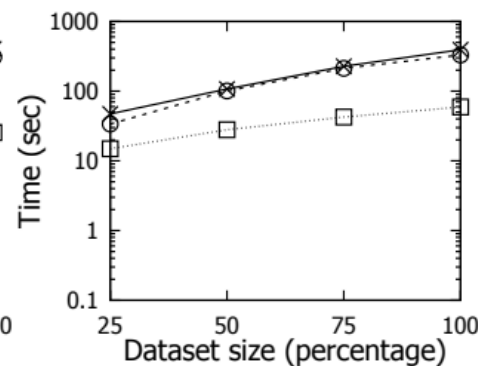
Name	n	$ V $	$ E $	Category
Seattle	543,969	11,383	37,472	Crime events
London	822,382	19,984	75,609	Traffic accidents
New York	1,542,738	41,307	115,021	Traffic accidents
Los Angeles	2,108,154	35,933	109,708	Crime events



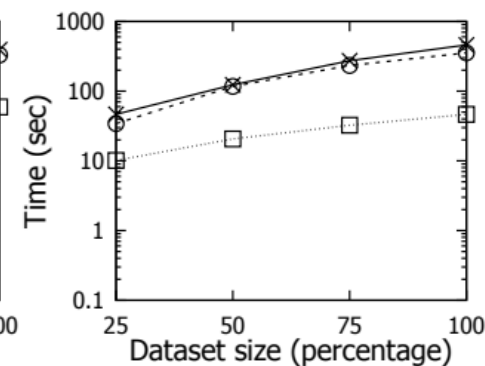
(a) Seattle



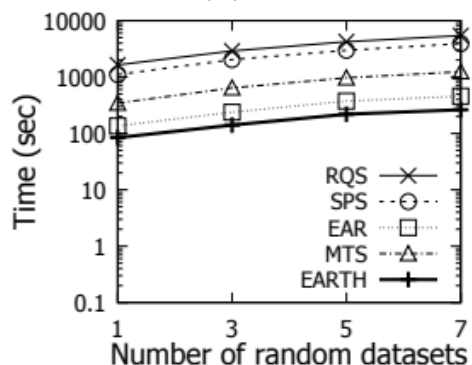
(b) London



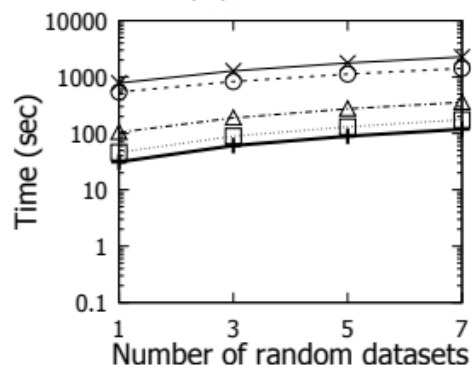
(c) New York



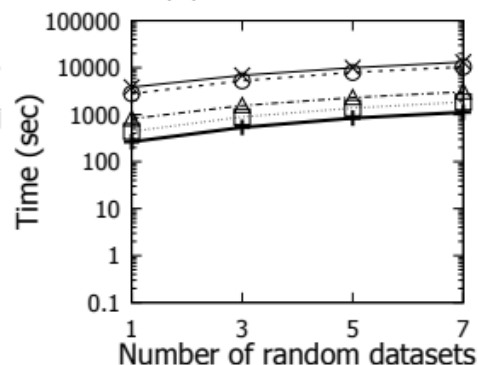
(d) Los Angeles



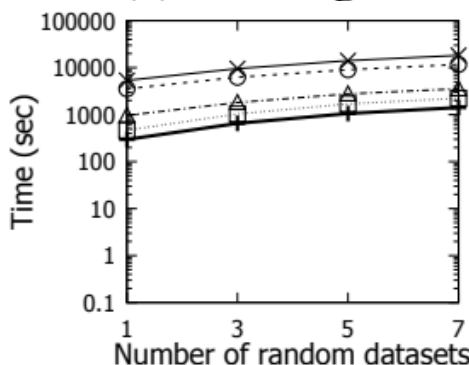
(a) Seattle



(b) London



(c) New York



(d) Los Angeles

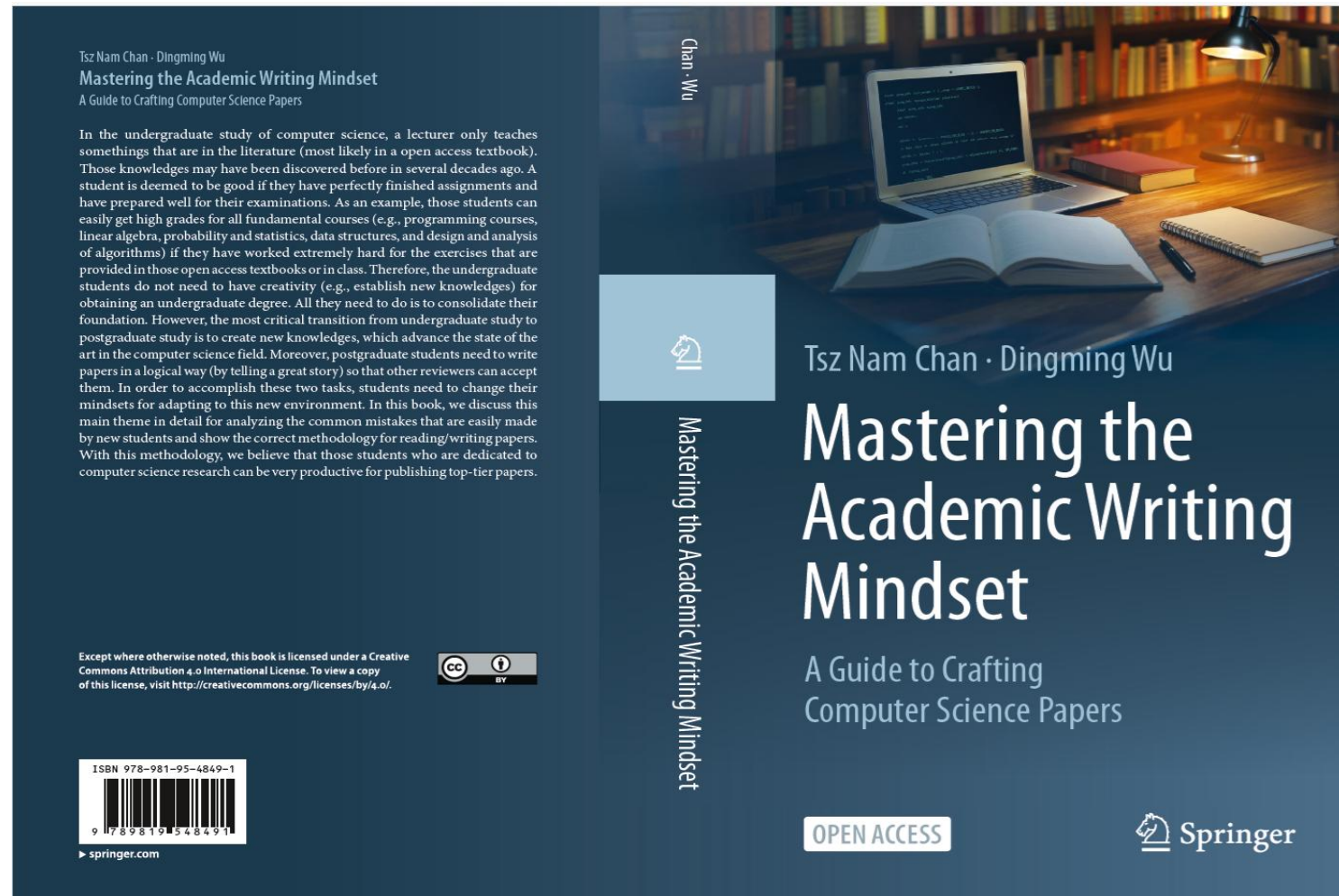
Future Work

- Further reduce the time complexity of spatiotemporal network K-function with a non-trivial approximation guarantee.
- Support other road network analysis tools.
 - Network inverse distance weighting (Network IDW)
 - Network Kriging
 - Network DBSCAN
- Add new functionalities in the FastNSA python library.
- Develop the FastNSA QGIS plugin.

Never Give Up!

- This paper has been rejected eight times before. ☹️
- SIGMOD 2024 (Rejected) → VLDB 2024 (Rejected) → ICDE 2024 (Rejected) → TKDE (Rejected) → SIGKDD 2025 (Rejected) → ICDE 2025 (Rejected) → VLDB 2026 (Rejected) → ICDE 2026 (Rejected) → **SIGKDD 2026 (Accepted with the Research Track Honorable Mention Award)**
- Keep submitting it to the best venue if you believe that this is a good work. 😊

Our New Book



<https://link.springer.com/book/10.1007/978-981-95-4850-7>